The `concept-factors` branch

From CoSpRo to Spatial Concept Distillation:
architectures, results and design rationale

Progress report

Abstract

This report describes the work on the `concept-factors` branch that followed the CoSpRo paper: the architectures it tested, the reasoning behind every design decision and the result of each experiment. The branch replaces CoSpRo's prior about *which* concept is spurious by a symmetric goal: the self-supervised student should encode every concept that CLIP/SpLiCE finds in the training images, so that a group-balanced linear probe can select the feature the task needs. Eight architectures were tested along three lines of work. Image-level concept distillation (a trained head, concept blocks and a cross-fitted ridge) doubles worst-group accuracy (WGA) on Spur-CIFAR10 and shows that a small dataset lets the student memorize any per-image target. Spatial Concept Distillation (SCD) supervises every location of the ResNet's 7×7 feature map with dense CLIP concept maps through the same cross-fitted ridge. It raises validation WGA over SimCLR by 6.1 (Waterbirds), 6.8 (MetaShift) and 22.5 points (Spur-CIFAR10) and beats its shuffled-map control on all three datasets; it is the main method of the branch. Four extensions on top of SCD aim to make the student separate an object from its usual context. Object compositing, which uses a text prior that types concepts as objects or scenes, reaches the highest Waterbirds WGA (59.1); the prior-free extensions match or fall below SCD. All numbers are validation numbers; the locked test runs are prepared.

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1 Summary of results

Table 1: Headline validation numbers (mean of the last four linear probes, epochs 425–500; ± is the standard deviation over student seeds; number of seeds in brackets). "Prior-free" means that no step of the method decides which concept is the class and which the spurious attribute.

	Waterbirds	MetaShift	Spur-CIFAR10	CelebA
SimCLR (reference)	46.7 ± 1.9 (6)	44.3 ± 7.3 (6)	15.6 ± 1.0 (2)	84.9 ± 0.0 (2)
CoSpRo (previous paper)	50.0 ± 3.2 (4)	42.9 ± 2.7 (10)	19.6 (1)	–
SCD, prior-free	52.8 ± 4.7 (6)	51.1 ± 5.4 (6)	38.1 ± 0.5 (2)	maps pending
same loss, shuffled maps	47.5 ± 2.5 (6)	45.1 ± 2.9 (6)	17.2 ± 2.0 (2)	
SCD + compositing, object prior	59.1 ± 7.1 (6)	52.3 ± 1.9 (4)	–	excluded

Main points.

1. CoSpRo helped on Waterbirds only and assumed that the spurious attribute is a removable context concept. The new branch drops that assumption and distils *all* concepts into the student.
2. Distilling concepts per image fails on small datasets because the student memorizes per-image targets. Distilling them *per location* of the feature map (SCD) works on all three datasets and beats a control that trains the same loss on another image's maps.
3. SCD makes the core feature stronger, while the student keeps the object tied to its usual context. The one extension that loosens this tie (compositing) needs an object-versus-scene prior.

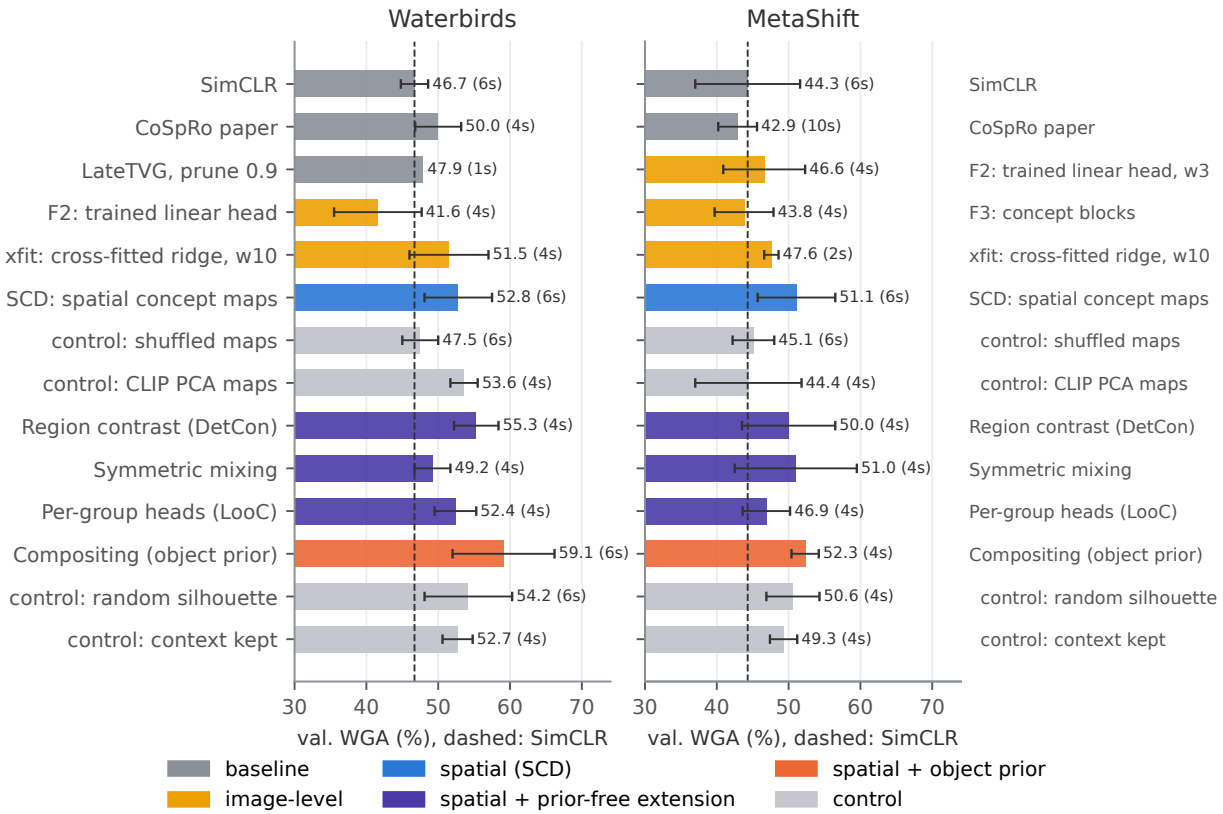


Figure 1: Every architecture of the branch on the two main datasets. Bars are validation WGA, whiskers one standard deviation over seeds, "(6s)" the number of seeds. Grey bars are controls. The dashed line is SimCLR. Section 9 discusses this figure row by row.

2 Problem setting and baselines

2.1 Spurious correlations and worst-group accuracy

Neural networks prefer the simplest feature that predicts the training labels, even when a harder feature is the one that generalizes (Geirhos et al., 2020; Shah et al., 2020). In Waterbirds the bird type agrees with the background on 95% of the training images (Sagawa et al., 2020). In MetaShift 88% of the training cats sit indoors and 88% of the dogs outdoors (Liang and Zou, 2022). A representation that learns "water" as evidence for "waterbird" or "indoor" as evidence for "cat" fails on the rare images that break the rule. Each image belongs to a group $g = (y, a)$ of class y and spurious attribute a. We report average accuracy and the accuracy of the worst group (Sagawa et al., 2020):

$$\mathrm{WGA} = \min_{g} \frac{1}{n_g} \sum_{i:\, g_i = g} \mathbf{1}\{\hat{y}_i = y_i\}. \tag{1}$$

Self-supervised learning inherits these correlations from its unlabeled images. Contrastive objectives can be met by the easiest discriminative feature and then suppress harder ones (Chen et al., 2021; Robinson et al., 2021); easy features dominate the representation even when a harder one is equally predictive (Hermann and Lampinen, 2020); image backgrounds alone carry much of the signal that ImageNet models use (Xiao et al., 2021a).

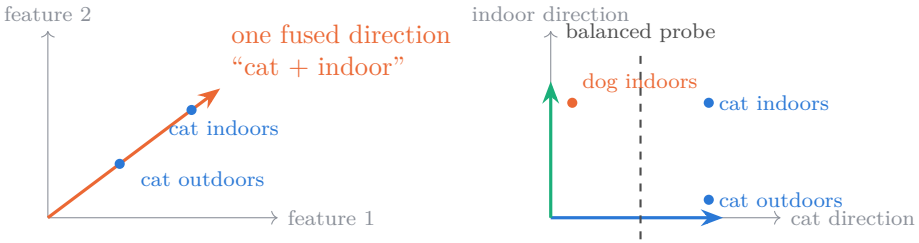
Table 2: Datasets. MetaShift and Spur-CIFAR10 served as development datasets; Waterbirds and CelebA received frozen configurations.

Dataset	Train images	Spurious structure and evaluation
Waterbirds (Sagawa et al., 2020)	4,795	bird type vs. land/water background; validation groups 467 / 466 / 133 / 133 images, the two waterbird groups smallest; test 642 per minority group
MetaShift (Liang and Zou, 2022)	1,700	cat/dog vs. indoor/outdoor, 88% agreement; 72 validation images per group, so WGA is noisy (one image = 1.4 points)
Spur-CIFAR10	45,000	CIFAR-10 classes (Krizhevsky, 2009); a 2-pixel coloured line agrees with the class on 95% of the training images; 100 (class, colour) groups
CelebA (Liu et al., 2015)	162,770	blond hair vs. gender; image-level runs only

2.2 Why a linear probe on a balanced set suffices: deep feature reweighting

Our evaluation fits a logistic regression on frozen features, the linear probe of representation analysis (Alain and Bengio, 2017), using a small *group-balanced* subset of the training split (224 images on Waterbirds, 400 on MetaShift, 1,400 on Spur-CIFAR10). Deep feature reweighting (DFR) showed that such a balanced last layer recovers high WGA whenever the representation encodes the core feature apart from the spurious one (Kirichenko et al., 2023). Izmailov et al. (2022) showed further that WGA after last-layer retraining depends mainly on the quality of the representation, and LaBonte et al. (2023) that a few hundred balanced examples suffice for the retraining. The quality of the representation is therefore the quantity to improve and the balanced probe measures it.

Goal derived from DFR. The representation must contain a *bird direction* and a *background direction* as two separate, linearly readable directions. It may keep the background; the balanced probe then puts its weight on the bird direction. The failure to prevent is the *fusion* of the two into one direction ("waterbird-and-water") or the *suppression* of the harder feature by the easier one (Figure 2).



fused: the probe cannot pick **separate: the probe splits on “cat”**

Figure 2: The geometry behind the goal. Left: a fused direction places cats outdoors close to the origin, so no linear probe separates them from dogs indoors. Right: with separate directions the balanced probe uses the cat axis alone.

2.3 The restriction: no prior about which concept is spurious

Label-free methods against spurious correlations rest on an assumption about what is spurious. LateTVG (Hamidieh et al., 2024) and LA-SSL (Zhu et al., 2023) assume that the spurious feature is the *easy* one, learned first or by the late layers. Background augmentation (Ryali et al., 2021), saliency-based copy-paste (Zhao et al., 2021) and object-aware contrastive learning (Mo et al., 2021) assume that it is the *background*. CoSpRo assumed a *removable context concept*. These assumptions hold on some datasets and fail on others: in CelebA the spurious attribute (gender) is a property of the person and lies in no background. The branch therefore adopts one restriction.

Restriction of the thesis method. No step of pretraining uses target labels, group labels or knowledge of which concept is the class and which the spurious attribute. Every concept enters the loss in the same way. Labels enter only the balanced probe and the post-hoc analysis, identically for every method.

2.4 What we compare against

- **SimCLR** (Chen et al., 2020) under the identical protocol: the reference for every gain.
- **A control for every new loss**: the same loss on another image’s targets (“shuffled”). It keeps every statistic of the targets and removes their link to the image content, so a gain above it is a gain of the concept content. This follows the control tasks of the probing literature, which separate what a representation encodes from what a flexible readout can fit (Hewitt and Liang, 2019). Most decisions of the branch rest on these controls.
- **CoSpRo** (our previous paper): 50.64 ± 1.97 test WGA on Waterbirds against 46.82 ± 3.65 for SimCLR; on MetaShift 42.9 ± 2.7 against 44.6 ± 6.9 over ten seeds.
- **LateTVG** (Hamidieh et al., 2024) reports 55.4 (Waterbirds) and 59.3 (MetaShift) WGA with SimCLR under its own protocol. Under our protocol one seed reaches 47.9 on Waterbirds; the matched six-seed test run is predeclared (`waterbirds_latetvg_test.yaml`).
- **LA-SSL** (Zhu et al., 2023) has an adaptation in the CoSpRo paper’s appendix.

Absolute accuracies are low (SimCLR reaches 52% average accuracy on Waterbirds) because a ResNet-18 trains from random initialization on 1,700 to 4,795 images. Like LateTVG and LA-SSL, the protocol pretrains on the dataset’s own images and measures what the SSL objective learns from them. Numbers with ImageNet-pretrained backbones (DFR reaches above 90% WGA on Waterbirds with a pretrained ResNet-50) answer a different question.

3 Starting point: CoSpRo

CoSpRo removed the subspace of one concept group from the CLIP embeddings, kept image pairs that became closer, filtered them (activation contrast, residual concept agreement, synthetic null groups) into a sparse teacher graph and transferred the graph to a SimCLR student through graph-aware batches and a relational KL loss in the spirit of relational distillation (Park et al., 2019). On Waterbirds it gained about four points of test WGA.

Two observations ended this line. First, the gain stayed on Waterbirds: CoSpRo scored below SimCLR on MetaShift over ten seeds and 19.6 against 15.6 on one Spur-CIFAR10 seed. Second, the method carries the prior that the spurious attribute is a removable context concept, the graph keeps wrong-class edges and the background stays linearly decodable at 91%.

4 Shared components of every architecture

4.1 Training and evaluation protocol

Table 3: Protocol shared by every run of the branch (inherited from CoSpRo so that numbers are comparable).

Student	ResNet-18 (He et al., 2016) from random initialization. Waterbirds, MetaShift, CelebA: 224-pixel input ("large" stem, 7×7 convolution with stride 2 and max-pooling). Spur-CIFAR10: 32-pixel input (CIFAR stem)
SSL	SimCLR (Chen et al., 2020), 500 epochs (CelebA 250), batch 128, temperature 0.05, SGD with learning rate 0.01 decayed by 0.1 at epochs 350, 400 and 450, weight decay 10^{-4}, crop scale $[0.2, 1]$, float16 autocast with a GradScaler (Micikevicius et al., 2018) on one V100
Probe	logistic regression (L2 coefficient 0.001) on the pooled 512-d features, fitted on the group-balanced subset of the training split, evaluated every 25 epochs
Reported value	mean of the last four probes (epochs 425 to 500); this damps single-probe noise on MetaShift's 72-image groups
Seeds	4 to 6 for Waterbirds and MetaShift, 2 for Spur-CIFAR10 and CelebA; differences are reported paired by seed (Bouthillier et al., 2021)
Split	validation; the locked test split stays unused until the configuration is final (Gulrajani and Lopez-Paz, 2021)

4.2 The ResNet-18 student and why the concept loss reads the 7×7 map

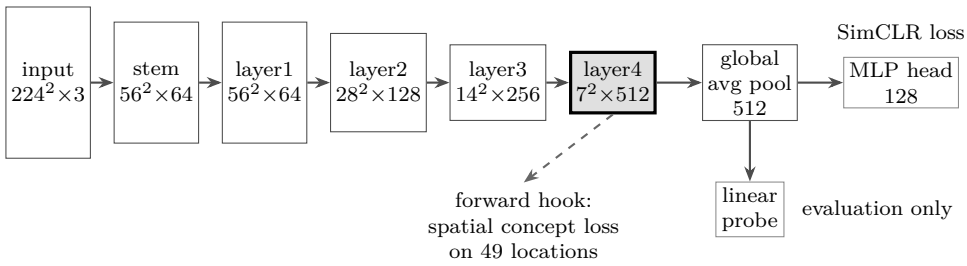


Figure 3: The student. SimCLR and the probe read the 512-d vector after global average pooling. The spatial concept loss reads the $7 \times 7 \times 512$ output of `layer4` through a forward hook, *before* pooling. Spur-CIFAR10 uses the CIFAR stem, so its map is $4 \times 4 \times 512$.

The network keeps every layer; the spatial loss taps the tensor that global average pooling would average. Four reasons fix this choice.

- **Pooling is where concepts fuse.** Global average pooling (Lin et al., 2014) turns the map into one 512-d vector per image, so a cat and a couch in the same image become one vector. Across images they co-occur (88%) and an image-level target separates "cat" from "couch" only on the few images where they disagree. In the 7×7 map they occupy different locations of *every* image. Class activation maps rest on the same property: the last convolutional map before pooling still localizes the objects that the pooled vector mixes (Zhou et al., 2016).
- **The last stage carries object and scene semantics and feeds the probe.** Units of the last convolutional stage respond to objects and scenes, earlier stages to textures and parts (Bau et al., 2017). `layer4` feeds the pooled vector directly, so shaping it shapes exactly the representation the probe evaluates. Regressing a teacher signal from an intermediate feature map through a small readout is the hint training of FitNets (Romero et al., 2015); dense SSL methods attach their losses to the same map (Hénaff et al., 2021; Wang et al., 2021; Xie et al., 2021).
- $7 = 224/32$**.** ResNet-18 halves the resolution five times (stem convolution, max-pooling, `layer2`, `layer3`, `layer4`); 49 locations remain.
- **A limitation built into this choice.** The theoretical receptive field of a `layer4` unit exceeds the 224-pixel view (Araujo et al., 2019); the effective receptive field is a smaller central region that still spans a large part of the view (Luo et al., 2016). A "cat" location therefore also sees the scene, which is why the student keeps part of the co-occurrence (Section 7.8).

4.3 SpLiCE concepts

A frozen OpenCLIP ViT-B/32 (LAION-2B weights; Radford et al., 2021) embeds every training image as z_i. SpLiCE (Bhalla et al., 2024) writes the centred, normalized embedding as a sparse non-negative combination of text directions from a vocabulary,

$$u_i = \mathrm{norm}(z_i - \mu), \qquad c_i \in \arg\min_{c \geq 0} \|u_i - D^\top c\|_2^2 + \lambda_1 \|c\|_1, \tag{2}$$

with L1 penalty 0.25. About nine concepts are active per image. MetaShift uses the LAION vocabulary (10,000 words), Waterbirds and Spur-CIFAR10 the Open Images v7 vocabulary (20,931 words). SpLiCE supplies a vocabulary of the concepts that occur in the training set, named and found without labels.

4.4 From concepts to concept factors: the meaning grouping

Summary. A vocabulary names one visual content several times (*cat, kitten, kitty*). A *concept factor* is a group of concepts that name one thing; its value on an image is the sum of its members' SpLiCE codes. The CoSpRo grouping rule, which required co-activation, left almost every concept alone and chained a bird to its background when its thresholds were lowered. The meaning grouping clusters concepts by raw CLIP text similarity with average linkage and a response veto. It works: it joins synonyms and keeps an object apart from its context on every dataset, because raw text embeddings place synonyms far closer than object–context pairs (Figure 4).

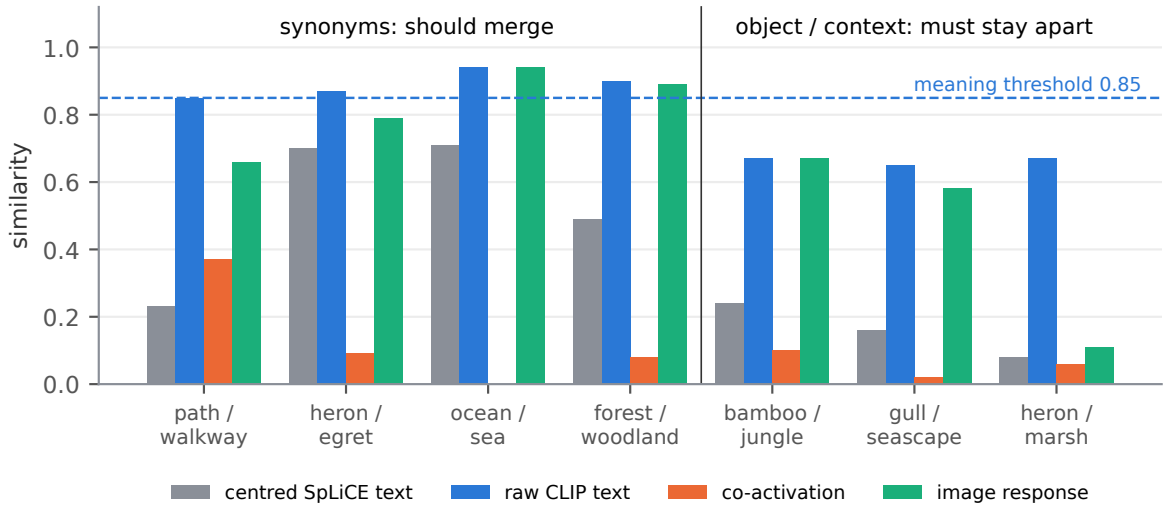


Figure 4: Four similarity signals on Waterbirds concept pairs. Co-activation (orange) fails for synonyms: the L1 penalty activates one of two synonyms per image (heron and egret co-activate at 0.09, ocean and sea at 0.00). Centred SpLiCE text directions (grey) also separate synonyms. Raw CLIP text embeddings (blue) put synonyms at 0.85 to 0.94 and object–context pairs at 0.64 to 0.72, so a threshold at 0.85 separates them.

Why co-activation fails. The L1 penalty of the lasso selects one variable out of a group of strongly correlated ones and sets the others to zero (Tibshirani, 1996; Zou and Hastie, 2005). SpLiCE inherits this behaviour: two synonyms are near-duplicate dictionary rows, so each image activates one of them. On Waterbirds 499 of 502 CoSpRo groups were single concepts. Lowering both thresholds linked concepts through chains of pairwise matches, the known failure of single linkage, which is what connected components of a pairwise graph compute (Murtagh and Contreras, 2012): one group held hummingbird, parrot, jungle and bamboo.

The procedure (`cospro/cli/build_meaning_groups.py`):

1. keep concepts active on 0.2% to 95% of the training images;
2. similarity $s_{k\ell} = \cos(t_k, t_\ell)$ of the raw CLIP text embeddings, set to -1 (veto) when the images' responses $u_i^\top d_k$ and $u_i^\top d_\ell$ correlate below 0.5;
3. average-linkage clustering cut at text similarity 0.85. Average linkage joins two clusters only when their members agree on average, which stops chains (Murtagh and Contreras, 2012);
4. a factor's text direction is the normalized mean of its members' text embeddings;
5. keep factors active on 2% to 90% of the images (5% to 90% for the image-level variants).

Examples: {kitty, kitten, cat}, {canine, dog}, {cozy, comfy, cosy}. Result: 242 factors on Waterbirds, 130 on MetaShift and 147 on Spur-CIFAR10 (2% band).

Why these numbers.

- **Text similarity 0.85** lies between synonyms (0.85–0.94) and object–context pairs (0.64–0.72); a sweep over 576 factor sets on MetaShift chose it together with the response veto 0.5.
- **Response veto 0.5**: two words with similar text whose images respond differently stay apart (heron and marsh: response correlation 0.11).
- **Frequency band 2%–90%**: a factor on fewer than 2% of the images occurs about once per half batch (64 images), so a fit on half a batch cannot learn it. At 5% the MetaShift factors lose

every scene concept (couch, lawn, bed, window, bench); at 2% they keep them. Factors above 90% carry no contrast between images.

5 Overview of the explored architectures

The architectures form three lines of work. Each line answers a question raised by the previous one: image-level targets show that concepts help and that small datasets are memorized; spatial targets remove the memorization and raise WGA on every dataset; the third line tries to separate the object from its usual context, which the spatial student leaves fused.

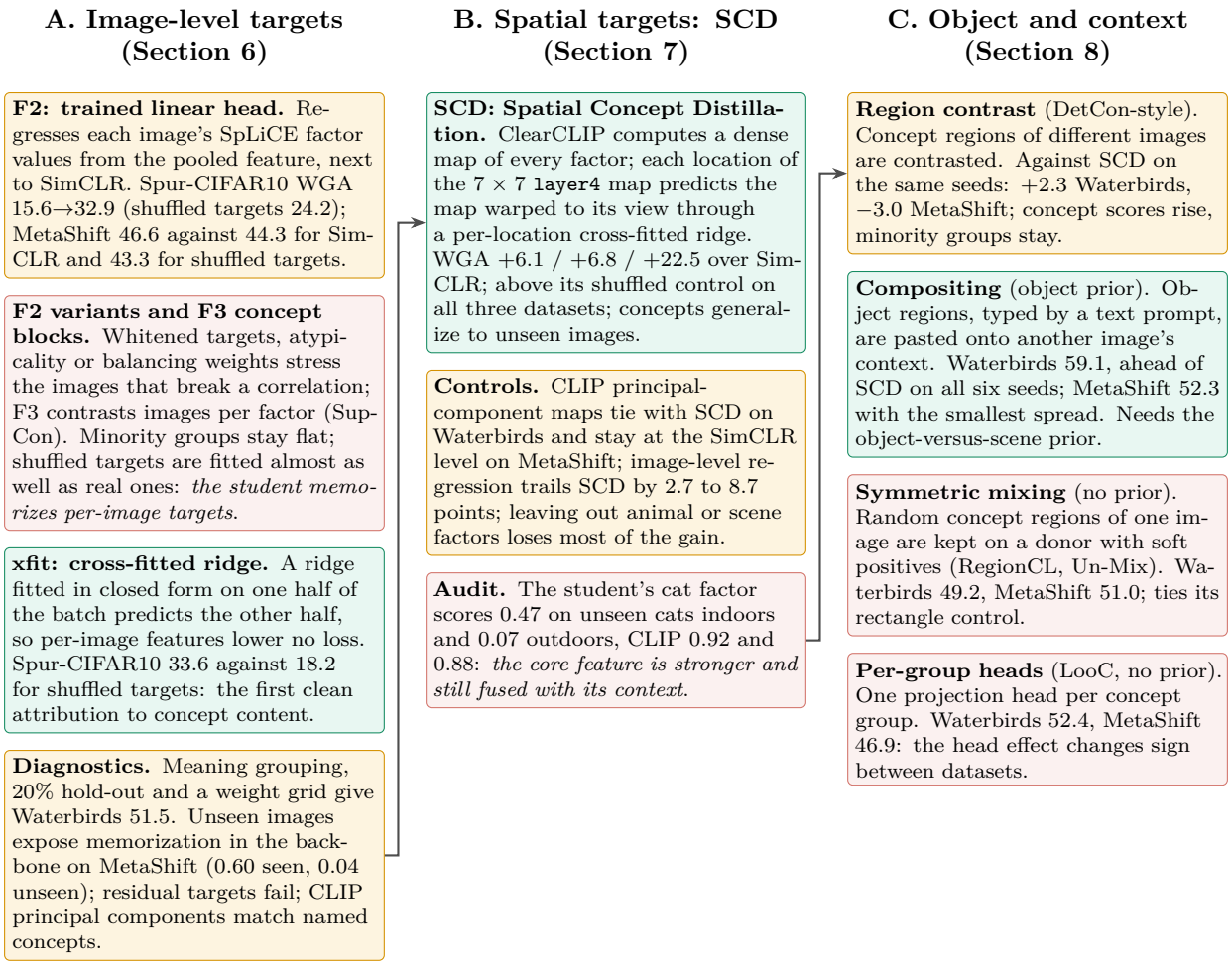


Figure 5: Map of the explored architectures. Green: kept; yellow: partial or informative; red: below the reference or explained away by a control. Numbers are validation WGA.

6 Image-level concept distillation

The methods of this section attach the concept targets to whole images: every image receives one vector of factor values, which the student predicts from its pooled feature. This is the simplest form of distilling a teacher's concepts (Hinton et al., 2015) and the natural first test of the goal of Section 4.4.

6.1 F2: factor distillation with a trained linear head

Summary. A linear head on the pooled feature regresses each image's SpLiCE factor values next to the SimCLR loss. It doubles WGA on Spur-CIFAR10 (15.6 to 32.9) and leaves MetaShift within the seed noise of SimCLR, with the gain in the majority groups. Reweighting and whitening variants change little. The cause is memorization: on 1,700 images the trained head fits shuffled targets almost as well as real ones, so the loss is met without learning concepts.

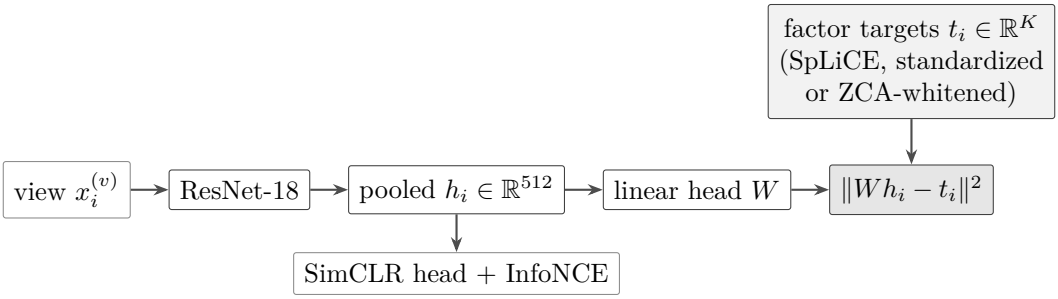


Figure 6: F2. A linear head predicts each image's factor values from the pooled backbone feature. The head is linear so that the factors must be linearly readable, as the probe reads them.

The loss is $\mathcal{L} = \mathcal{L}_{\text{SimCLR}} + \lambda \frac{1}{2B} \sum_{i,v} \|Wh_i^{(v)} - t_i\|^2 / K$, with the weight λ switched on at epoch 10 after a 10-epoch linear warm-up. Three variants target the images that break a correlation:

- **Whitened targets** (ZCA, ridge 0.1; Kessy et al., 2018) keep for each factor the part the other factors leave unexplained. This part is large exactly on the images that break a correlation.
- **Atypicality weights** scale each image's loss by its mean squared whitened factor value: images whose concepts break the dataset's correlations receive more weight (minority images about twice the weight on a synthetic fixture). The idea is the upweighting of hard examples in JTT (Liu et al., 2021), computed from concepts.
- **Balancing weights** are per-image weights that minimize the squared correlations between factor presences, so that under them "indoor" no longer predicts "cat". This is the sample reweighting of stable learning (Shen et al., 2020; Zhang et al., 2021).

Table 4: F2 results (validation WGA / average accuracy; groups on MetaShift: cat-in / cat-out / dog-in / dog-out).

Dataset	Arm	WGA	Acc.	Groups
Spur-CIFAR10	SimCLR	15.6 ± 1.0	59.5	
	F2, standardized	32.9 ± 1.6	69.5	
	F2, shuffled targets	24.2 ± 0.4	58.6	
MetaShift	SimCLR	44.3 ± 7.3	55.2	57 / 47 / 53 / 64
	F2, weight 3	46.6 ± 5.7	61.5	70 / 47 / 54 / 74
	F2, weight 3, shuffled	43.3 ± 8.1	54.8	57 / 45 / 53 / 65
	F2, whitened, weight 3	47.9 ± 6.4	56.9	57 / 51 / 55 / 65
	F2, atypicality weights	50.5 ± 3.8	58.8	59 / 52 / 55 / 70
	F2, balancing weights	44.2 ± 8.1	59.0	64 / 44 / 55 / 72
Waterbirds	F2 (first frozen configuration)	41.6 ± 6.1	51.2	below SimCLR

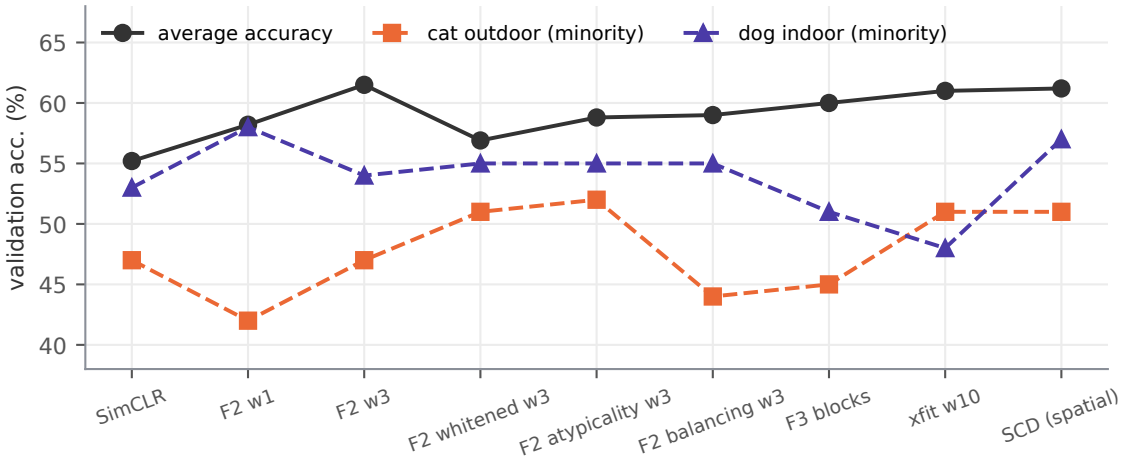


Figure 7: MetaShift across the image-level variants. Average accuracy rises with almost every concept loss; the two minority groups (cat outdoors, dog indoors) stay within a few points of SimCLR. The gain goes to the majority groups.

6.2 F3: concept blocks (supervised contrast per factor)

Summary. One small embedding block per factor applies a supervised contrastive loss in which two images that both show the factor are positives, weighted towards pairs whose other concepts differ. The idea is that a fused "cat-and-indoor" direction cannot rank "cat on street" above "dog on sofa" inside the cat block. On MetaShift it scores 43.8 WGA, below its own controls and the loss reaches the same value with shuffled concepts: the blocks are met by memorization as well.

Method. The 32 most balanced factors each get a linear block. Inside block k, images that both show factor k are positives under SupCon (Khosla et al., 2020); a positive pair weighs more when the two images' other factors differ ("cat on sofa" with "cat on street"). The backbone would then need a cat direction that survives a change of context. **Result:** 43.8 ± 4.1 WGA; the control with uniform pair weights scored 49.7 ± 2.5 and the shuffled control 46.4 ± 3.5.

6.3 Diagnosis: the student memorizes per-image targets

The controls explain every MetaShift result. F2 at weight 3 explains 0.41 of the real target variance and 0.33 of the shuffled one; the F3 block loss ends at 3.34 with real concepts and 3.37 with shuffled ones. Deep networks fit random labels on small datasets (Arpit et al., 2017; Zhang et al., 2017) and a SimCLR backbone is built to tell individual images apart (Wang and Isola, 2020; Wu et al., 2018). A per-image target on 1,700 images can therefore be read off the instance identity that SimCLR provides anyway. Any auxiliary loss attached to single images is then met without learning concepts. On Spur-CIFAR10 (45,000 images) shuffled targets reach only 0.02 explained variance, so the concepts shape the student there. The remedy has to remove the reward for memorizing an image; the cross-fit of the next section does exactly that.

6.4 xfit: the cross-fitted ridge

Summary. The trained head is replaced by a ridge regression fitted in closed form on one half of the batch and evaluated on the other half. A feature that only identifies one image carries nothing to the other half, so memorization lowers no part of the loss. On Spur-CIFAR10 xfit matches F2 (33.6 WGA) while its shuffled control falls to the SimCLR level: the whole gain comes from the concept content. On MetaShift it removes the memorization of the head and leaves WGA at the SimCLR level, because the backbone still stores the targets of its training images (Section 6.5).

Idea. Fitting an auxiliary model on one part of the sample and evaluating it on another is the sample splitting (cross-fitting) of double machine learning (Chernozhukov et al., 2018). We apply it to a training loss: only directions that transfer from half A to half B lower the error. The ridge is solved in closed form and differentiated through, as in the closed-form heads of R2D2 (Bertinetto et al., 2019) and MetaOptNet (Lee et al., 2019), so the backbone receives gradients through the solve.

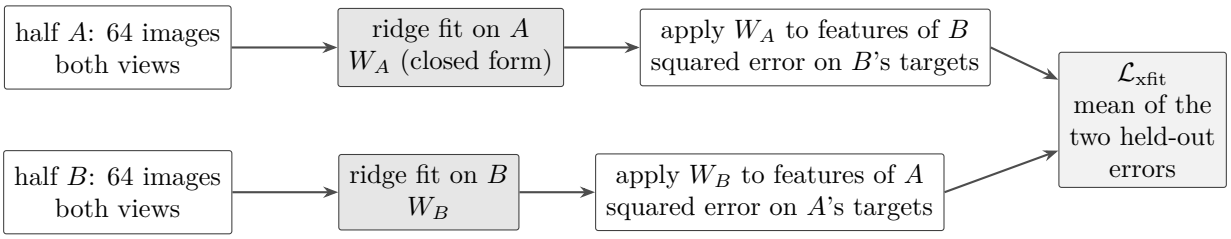


Figure 8: Cross-fitting. Gradients flow through the solve and the prediction into the backbone. Both views of an image stay in the same half, so no half sees a view of an image it predicts.

With unit-norm centred features H, targets T and per-image weights D, the dual (kernel) form of the ridge (Hoerl and Kennard, 1970; Saunders et al., 1998) is

$$\hat{T}_B = H_B H_A^\top \left(D_A H_A H_A^\top + \rho I\right)^{-1} D_A T_A, \tag{3}$$

one $n \times n$ solve per half (code: `cross_fit_predictions` in `cospro/methods/concept_factors.py`).

Why unit-norm rows and an absolute ridge $\rho = 1$. The first version scaled the ridge with the feature norm, which made it small. Fits of 64 images in 512 dimensions then nearly interpolated, the solve became ill conditioned, three of four Spur-CIFAR10 runs produced NaN features and the held-out explained variance was negative everywhere. Normalizing every feature row stops the backbone from weakening the ridge by shrinking its features and an absolute ridge keeps every eigenvalue of the system at least ρ (Hoerl and Kennard, 1970). Ridge 10 performed no better on MetaShift (42.5 WGA).

Table 5: Cross-fit with unit-norm rows and $\rho = 1$ (seeds 1 and 2). Batch EV: held-out explained variance of the in-batch fit.

Dataset	Arm	WGA	Acc.	Batch EV
Spur-CIFAR10	SimCLR	15.6 ± 1.0	59.5	
	F2 trained head	32.9 ± 1.6	69.5	0.10
	xfit	33.6 ± 2.3	69.7	0.13
	xfit, shuffled	18.2 ± 3.2	57.5	-0.02
MetaShift	xfit (145 factors)	43.9 ± 3.2	54.8	-0.01
	xfit, shuffled	41.7 ± 0.0	54.4	-0.03

6.5 What the student can learn: bounds, weights and unseen images

Learnability bound. A ridge fitted on 64 images can only predict factors that occur often enough. Running the same fit on frozen CLIP features gives an upper bound: with 278 factors from 1% frequency even CLIP scores -0.015; with the 24 factors from 5% it scores 0.16 (shuffled: -0.07). Hence the 5% band for image-level runs.

Loss weight. At weight 10 the 24 MetaShift factors reach 0.72 to 0.87 explained variance on the training images, *three times the CLIP score*, while WGA stays at 45.5. A probe on the 24 factor values themselves reaches 77% WGA. The student stores something other than the concepts.

The hold-out diagnostic. A fixed fifth of the training images gets loss weight 0 in the concept loss (SimCLR still trains on them). A ridge fitted on the other images then scores the factors on these *unseen* images. This exposes memorization in the backbone, which the cross-fit cannot see: the cross-fit removes the reward for memorizing inside one batch, while the backbone can still store each training image's targets across epochs.

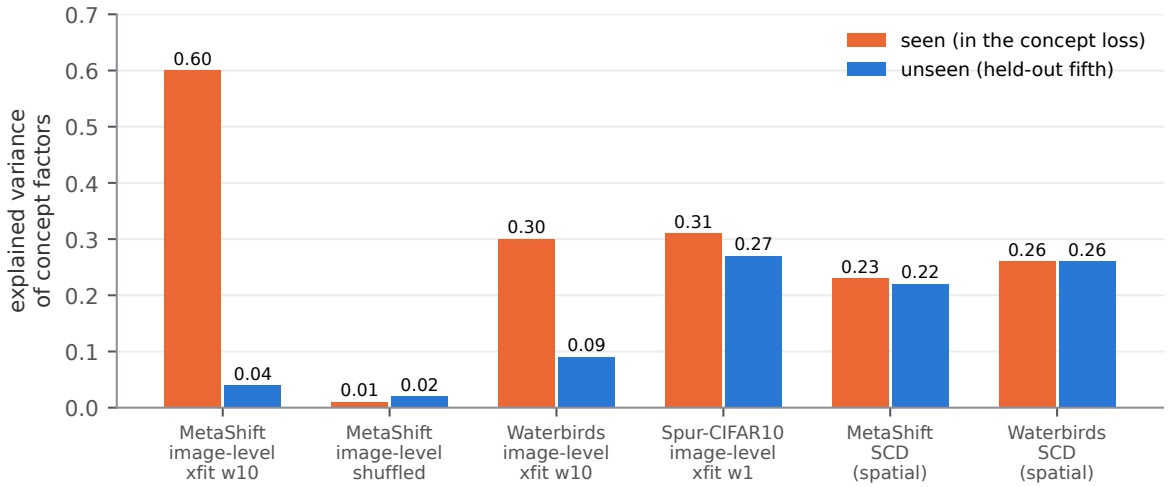


Figure 9: Explained variance of the factors on images inside the concept loss (seen) and outside it (unseen). Image-level xfit on MetaShift: 0.60 seen, 0.04 unseen; the backbone stores the targets of its 1,700 images. Spur-CIFAR10 (45,000 images) generalizes: 0.31 against 0.27. SCD (spatial, scored on dense responses) shows no gap on either dataset.

6.6 Waterbirds with the frozen image-level configuration

The configuration chosen on MetaShift (meaning factors from 5%, 70 Open Images factors, weight 10, hold-out) reaches 55.3 WGA on the first two Waterbirds seeds and 51.5 ± 5.5 on four, with accuracy 57.5 against 52.3 for SimCLR and 41.4 ± 3.0 for its shuffled control. The same configuration on Spur-CIFAR10 drops to 22.7 (weight 1 gives 33.6): at weight 10 even 45,000 images are memorized. The weight grid on Spur-CIFAR10 shows that the unseen score cannot choose the weight label-free (0.32 to 0.33 for weights 1 to 10 while WGA moves within noise).

6.7 Two final diagnostics

Decoupling score. Each factor minus its linear regression on the other factors (its *residual*) is the part a fused direction cannot predict. The student explains 0.03 to 0.04 of the residuals against 0.15 to 0.17 for CLIP. Training directly on residual targets fails: the student cannot predict them and Waterbirds WGA falls to 41.7 ± 0.9, the level of the shuffled control.

CLIP control. Distilling as many principal components of the CLIP image embeddings as there are factors gives the same or better numbers (Waterbirds 51.0, MetaShift 54.2, Spur-CIFAR10 36.3 to 37.0). At the image level the concept names add nothing over generic CLIP information, in line with plain feature distillation from CLIP (Yang et al., 2024).

6.8 Outcome of the image-level line

Image-level concept distillation helps when the student can generalize the targets (Spur-CIFAR10: 15.6 to 33.6; Waterbirds: 46.7 to 51.5) and the cross-fit makes the attribution clean. Three findings limit it: small datasets are memorized in the backbone; the student learns the part the factors share, which is the dataset's co-occurrence (Träuble et al., 2021); named concepts match CLIP principal

components. Image-level targets can separate "cat" from "couch" only through the images where the two disagree. That limit motivated the spatial loss.

7 Spatial Concept Distillation (SCD)

Summary. SCD tells the student *where* each concept is. A frozen CLIP model computes a dense map of every concept factor over every training image; during SimCLR training every location of the student's 7×7 feature map predicts the concept values at its place in the view, through the cross-fitted ridge of Section 6.4. SCD works: it is the first method of the project that beats its shuffled control on every dataset tested (WGA +6.1, +6.8 and +22.5 over SimCLR on Waterbirds, MetaShift and Spur-CIFAR10) and its concepts generalize to images outside the loss. It works because location separates concepts that co-occur in images: cat and couch share an image label but occupy different cells, so the student must encode both. Its limit is that each cell sees most of the view, so the student still predicts the cat partly from the scene. SCD is the prior-free method of the thesis; this section describes it in the form of the paper draft (`paper/SpatialConcepts.tex`).

7.1 Motivation

SSL representations inherit the co-occurrences of their unlabeled training images: when cats appear mostly indoors, an encoder trained from scratch learns indoor scenes as evidence for cats. Deep feature reweighting shows that a linear head retrained on a small group-balanced set recovers WGA whenever the representation encodes the core feature apart from the spurious one (Kirichenko et al., 2023). A frozen vision-language model supplies semantic structure learned from external data and SpLiCE exposes it as sparse non-negative coordinates associated with text concepts (Bhalla et al., 2024). The image-level experiments show that these coordinates help when the student can generalize them and fail where the concepts co-occur. We hypothesize that the *location* of a concept is the missing signal: at the image level cat and couch appear together, while at the pixel level they occupy different places in every image.

7.2 Related work

Dense and region-level self-supervised learning. SimCLR contrasts two views of an image against the other images in the batch (Chen et al., 2020) and pools the feature map before its projection head. VADeR and DenseCL add a contrastive loss between matched locations of the two views (Pinheiro et al., 2020; Wang et al., 2021) and PixPro asks locations that cover the same pixels to agree (Xie et al., 2021). DetCon pools the feature map inside unsupervised masks and contrasts the pooled regions (Hénaff et al., 2021). These methods supervise locations with the student's own features or with unnamed segments. SCD supervises every location with named concept values from a frozen teacher and keeps the SimCLR objective unchanged.

Dense concepts from vision-language models. CLIP supplies a shared image–text space (Radford et al., 2021). MaskCLIP shows that its patch tokens align with text once the last attention pooling is bypassed (Zhou et al., 2022). SCLIP and ClearCLIP obtain cleaner maps by changing the attention of the last block (query–query or self–self correlations) and ClearCLIP also removes its residual and feed-forward branches (Lan et al., 2024; Wang et al., 2024). SCD uses the ClearCLIP construction offline, on the training images only.

Distillation into a student. Knowledge distillation trains a student to match a teacher's outputs (Hinton et al., 2015); FitNets regresses the teacher's signal from an intermediate feature map of the student (Romero et al., 2015). Feature distillation from CLIP works with plain feature mimicry (Yang

et al., 2024) and AM-RADIO distils several teachers with summary and spatial losses (Ranzinger et al., 2024). SCD distils a low-dimensional concept map through a regression that is refitted in every batch, in the manner of differentiable closed-form solvers in meta-learning (Bertinetto et al., 2019; Lee et al., 2019).

Spurious features in SSL and language-guided attention. LA-SSL (Zhu et al., 2023) and LateTVG (Hamidieh et al., 2024) assume that the spurious feature is the easy one; background augmentations (Ryali et al., 2021) assume that it is the background. GALS grounds a language description of the class in CLIP attention and guides a supervised network towards those pixels (Petryk et al., 2022), a descendant of explanation constraints from annotated masks (Ross et al., 2017). SCD shares the use of language-grounded locations with GALS, while it names no class, uses every concept and trains a separate encoder without labels.

7.3 Method

SCD has four stages: encode images as sparse named concepts; group the concepts into factors that name one thing each; map every factor densely over every training image; train a SimCLR student whose feature map predicts the warped maps of its views. The frozen branch runs once on the training images and provides the student with the location of every concept, without target or spurious information (Figure 10).

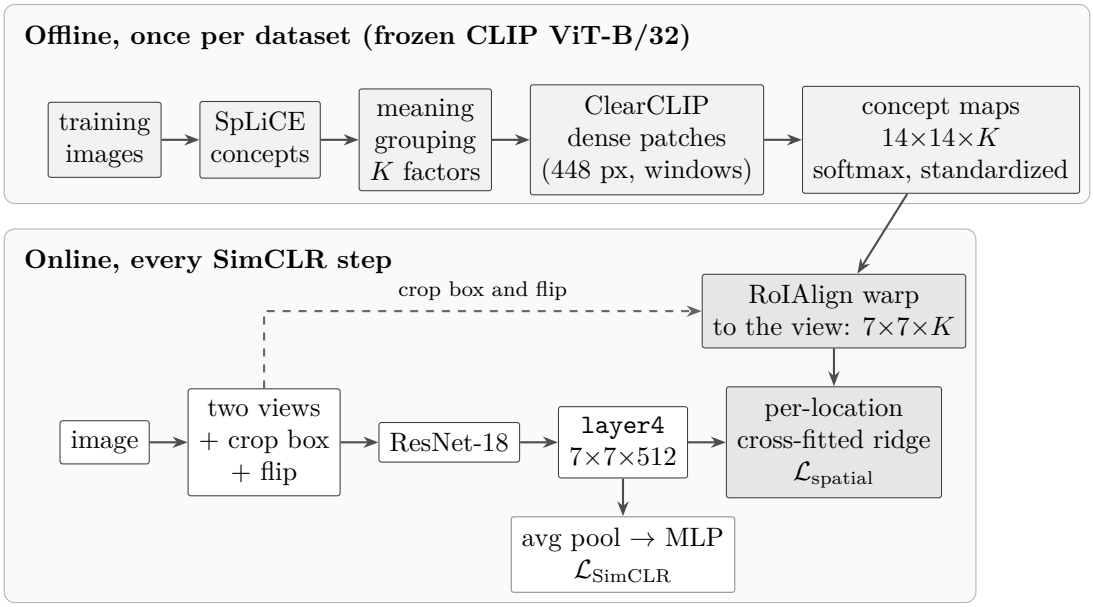


Figure 10: SCD. The teacher branch runs once (40 seconds on one GPU for MetaShift) and stores the maps. The student branch adds one loss to unchanged SimCLR. At inference only the student backbone and the probe remain; teacher, maps and heads are discarded.

Stages 1 and 2: concepts and concept factors. SpLiCE codes c_i (Section 4.3) give the vocabulary of the training set. The meaning grouping (Section 4.4) turns it into K factors, each with a text direction $d_F = \text{norm}\left(\frac{1}{|F|}\sum_{k\in F}\text{norm}(t_k)\right)$ and an image-level activation $a_{iF} = \sum_{k\in F} c_{ik}$. Waterbirds has 242 factors, MetaShift 130 and Spur-CIFAR10 147.

Stage 3: dense concept maps. Plain CLIP patch tokens of the last layer align poorly with text, because the final attention mixes global information into every token (Zhou et al., 2022). Following ClearCLIP (Lan et al., 2024), the last transformer block replaces its query–key attention by query–query attention and drops its residual and feed-forward branches. With X the normalized

tokens entering this block, the patch embeddings in the text space are

$$P = \mathrm{norm}\Big(\mathrm{LN}\left(\mathrm{softmax}(QQ^\top/\sqrt{d_h})\,VW_O\right)W_{\mathrm{proj}}\Big), \qquad Q = XW_Q,\ V = XW_V, \tag{4}$$

applied per attention head and normalized per patch. Images are read at 448 pixels through 224-pixel windows with a stride of 112 pixels (Figure 11); window scores are averaged where windows overlap and pooled to a 14×14 grid. At every cell u the map is a softmax over the factors,

$$M_{iF}(u) = \frac{\exp(\cos(p_i(u), d_F)/T)}{\sum_G \exp(\cos(p_i(u), d_G)/T)}, \qquad \tilde{M}_{iF} = (M_{iF} - m_F)/s_F, \tag{5}$$

with temperature $T = 0.01$, so each cell distributes one unit of mass over the factors. Each factor's map is then standardized over all images and cells (m_F, s_F) and stored in float16.

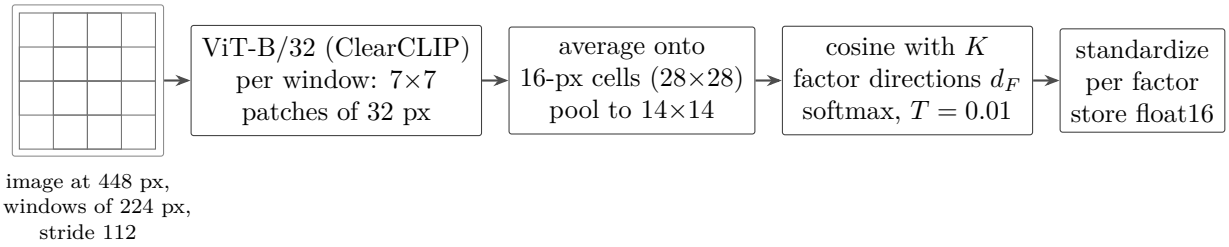


Figure 11: Map construction. Code: `cospro/pipeline/concept_maps.py`; launcher: `scripts/build_concept_maps.sbatch`.

Stage 4: warping the maps onto each view. The standard two-view SimCLR objective remains unchanged. The SSL transform records the crop box and flip of every view $x_i^{(v)}$. RoIAlign (He et al., 2017) cuts the view's box out of $\tilde{M}_i$ (a 2×2 sampling grid per output bin), resizes it to the student's feature map and flips it with the view, giving targets $Y_i^{(v)} \in \mathbb{R}^{K \times h \times w}$ that describe exactly the pixels the view shows. The feature map $H_i^{(v)} \in \mathbb{R}^{C \times h \times w}$ is the output of `layer4` before global pooling ($7 \times 7 \times 512$ for 224-pixel views).

Stage 4: the per-location cross-fitted ridge. Every location u gives a unit-norm feature row $r = \mathrm{norm}(H_i^{(v)}(u)) \in \mathbb{R}^{512}$ and a target $y = Y_i^{(v)}(u) \in \mathbb{R}^K$. A trained linear head on these rows could store a target per image, as the image-level heads did. We therefore split the images of a batch into halves A and B, with both views of an image in one half and fit a ridge regression on the locations of A in closed form,

$$W_A = \left(\bar{R}_A^\top \Omega_A \bar{R}_A + \rho I\right)^{-1} \bar{R}_A^\top \Omega_A (Y_A - \bar{y}_A), \tag{6}$$

where $\bar{R}_A$ stacks the rows of A centred by their weighted mean $\bar{r}_A$, Ω_A holds the per-view loss weights and $\rho = 1$ is an absolute ridge. The fit predicts every location of B, $\hat{Y}_B = (R_B - \bar{r}_A)W_A + \bar{y}_A$, and the reverse fit predicts A. The loss is the held-out squared error,

$$\mathcal{L}_{\mathrm{spatial}} = \frac{1}{2} \sum_{(S,S') \in \{(A,B),(B,A)\}} \frac{\sum_{v \in S'} \omega_v \frac{1}{hwK} \left\| \hat{Y}_v - Y_v \right\|_2^2}{\sum_{v \in S'} \omega_v}. \tag{7}$$

Here the primal form is cheaper than the kernel form of Section 6.4: one 512×512 solve over $64 \cdot 2 \cdot 49 = 6{,}272$ rows. The solve runs in float32 outside autocast. Gradients pass through the solve and the prediction. A feature that identifies one image carries nothing to the other half; only concept directions shared across images lower the loss.

A fixed fifth of the training images carries loss weight $\omega = 0$: SimCLR still trains on them and they measure whether the student carries concepts to images outside the concept loss. The complete objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{SimCLR}} + \lambda(t)\,\mathcal{L}_{\mathrm{spatial}}, \qquad \lambda(t) = 0 \text{ up to epoch 10, linear to } \lambda_{\max} = 5 \text{ at epoch 20.} \tag{8}$$

One training step.

1. Draw 128 images, two augmented views each; record each view's crop box and flip.
2. Run the student; keep the pooled vector for SimCLR and the $7 \times 7 \times 512$ `layer4` map from the hook.
3. Warp each image's stored $14 \times 14 \times K$ map to each view's box and flip: $7 \times 7 \times K$ targets.
4. Split the images into halves; fit the ridge on all locations of one half, predict the other, both directions.
5. Add $\lambda(t)$ times the held-out error to the SimCLR loss and back-propagate through the solve.

7.4 Design choices and their justification

- **ClearCLIP as the dense teacher.** Training-free dense CLIP methods give patch–text alignment without any segmentation labels (Lan et al., 2024; Wang et al., 2024; Zhou et al., 2022); ClearCLIP reports the highest open-vocabulary segmentation scores of the three and keeps the restriction (no labels).
- **448 pixels through 224-pixel windows with stride 112.** ViT-B/32 sees a 224-pixel input as only 7×7 patches of 32 pixels, too coarse to separate a cat from its cushion. Sliding-window inference with 224-pixel windows and stride 112 on an enlarged image follows the evaluation of SCLIP and ClearCLIP (Lan et al., 2024; Wang et al., 2024); it gives an effective 16-pixel grid while each window stays at the resolution CLIP was trained on.
- 14×14**, twice the student's** 7×7**.** A SimCLR crop covers 20% to 100% of the image; a 20% crop still spans about 6×6 map cells, so RoIAlign has real content to resample into the 7×7 target.
- **Softmax with** $T = 0.01$**.** CLIP's learned logit scale is 100 (Radford et al., 2021), which is a temperature of 0.01; its zero-shot classifier applies this scale to the same cosines. Cosines of a patch with different text directions differ by a few hundredths and $T = 0.01$ turns them into a peaked distribution, so each cell mostly picks a winner.
- **Standardized per factor.** Every factor then has unit variance, so frequent factors (blanket, lawn) do not dominate the squared error.
- **RoIAlign for the warp.** RoIAlign samples the map at exact fractional coordinates with bilinear interpolation (He et al., 2017); it keeps the map aligned with the view's pixels for any crop box.
- **Cross-fit in place of a trained head**: the lesson of the image-level line, where a trained head stored per-image targets on small datasets.
- $\lambda_{\max} = 5$: from the weight grid 1, 3, 5 and 10 of the image-level loss; spatial weight 10 adds nothing (Waterbirds 53.5, MetaShift 48.3, two seeds).
- **Start at epoch 10 with a 10-epoch warm-up**: SimCLR forms features before the concept loss acts; inherited from F2, where starting at epoch 0 changed nothing.
- **20% hold-out**: introduced as a diagnostic. The arm with the hold-out scored higher than the arm without it (Waterbirds 52.8 vs. 49.5, MetaShift 51.1 vs. 46.7, six seeds) and became the main arm. The difference lies within the seed spread and has no explanation yet.
- **Batch 128**: 6,272 rows per half against 512 dimensions keep the fit well posed.

7.5 Experimental protocol

We evaluate on Waterbirds, MetaShift cats and dogs and Spur-CIFAR10. Each dataset gets its own concept factors and maps from its training images. We pretrain a randomly initialized ResNet-18 for

500 epochs with batch size 128, the same augmentations and optimization schedule for every method and SimCLR temperature 0.05. The concept loss uses ridge $\rho = 1$, weight 5 and the 20% hold-out. We report six student seeds on Waterbirds and MetaShift and two on Spur-CIFAR10. Frozen student features are evaluated every 25 epochs with the same logistic probe, fitted on the group-balanced subset of the training split. Each value is the mean of the last four probes (epochs 425 to 500). We report mean and sample standard deviation across student seeds and paired differences per seed (Bouthillier et al., 2021).

7.6 Results

Table 6: SCD against SimCLR and the shuffled-map control. Waterbirds and MetaShift: seeds 1–6; Spur-CIFAR10: seeds 1–2. Validation, mean of the last four probes.

	Waterbirds		MetaShift		Spur-CIFAR10	
Method	Avg.	WGA	Avg.	WGA	Avg.	WGA
SimCLR	52.4	46.7 ± 1.9	55.2	44.3 ± 7.3	59.6	15.6 ± 1.0
Shuffled maps	52.8	47.5 ± 2.5	56.1	45.1 ± 2.9	55.9	17.3 ± 2.0
SCD	**60.3**	52.8 ± 4.7	**61.2**	51.1 ± 5.4	**69.3**	38.1 ± 0.5

SCD beats SimCLR by 6.1 (Waterbirds), 6.8 (MetaShift) and 22.5 (Spur-CIFAR10) points of WGA and by 7.9, 6.0 and 9.8 points of average accuracy. It is ahead on five of six paired seeds on Waterbirds and MetaShift and on both Spur-CIFAR10 seeds (Table 7). The shuffled control stays within one point of SimCLR on Waterbirds and MetaShift and 1.6 points on Spur-CIFAR10, so the gain comes from the content of each image's own maps. The gain extends to the minority groups: on MetaShift cats outdoors rise from 47.0 to 51.2 and dogs indoors from 52.8 to 57.2; on Waterbirds landbirds on water rise from 53.6 to 65.1 (Appendix B).

Table 7: Paired SCD differences in percentage points per seed, mean of the last four probes.

	Waterbirds				MetaShift			
	vs. SimCLR		vs. shuffled maps		vs. SimCLR		vs. shuffled maps	
Seed	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA	ΔAvg.	ΔWGA
1	+9.30	+12.29	+3.50	+7.07	+9.72	+9.72	+7.99	+17.36
2	+11.45	+7.34	+10.15	+3.59	+3.56	+5.90	+4.60	+8.68
3	+6.80	+3.32	+7.09	+5.84	+13.54	+22.57	+4.69	+4.86
4	+4.05	+5.97	+9.47	+8.69	+2.86	−5.21	+9.11	+0.00
5	+11.41	+10.47	+10.36	+10.57	+6.51	+4.17	+1.48	+2.08
6	+4.30	−2.78	+4.61	−4.18	−0.26	+3.47	+2.60	+3.13
Mean	+7.88	+6.10	+7.53	+5.26	+5.99	+6.77	+5.08	+6.02

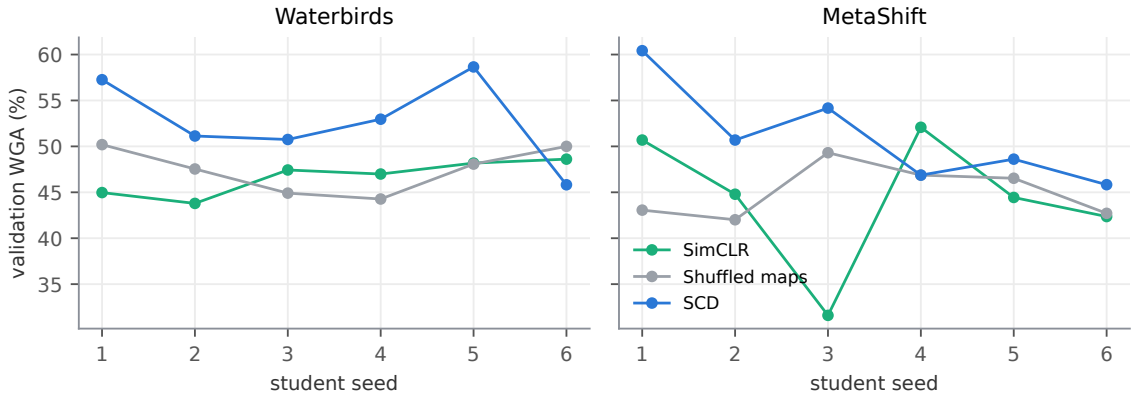


Figure 12: Per-seed WGA. MetaShift seeds spread by more than 20 points for SimCLR (31.6 to 52.1), which is why we report six seeds and paired differences. Seed 6 is hard for every method on Waterbirds.

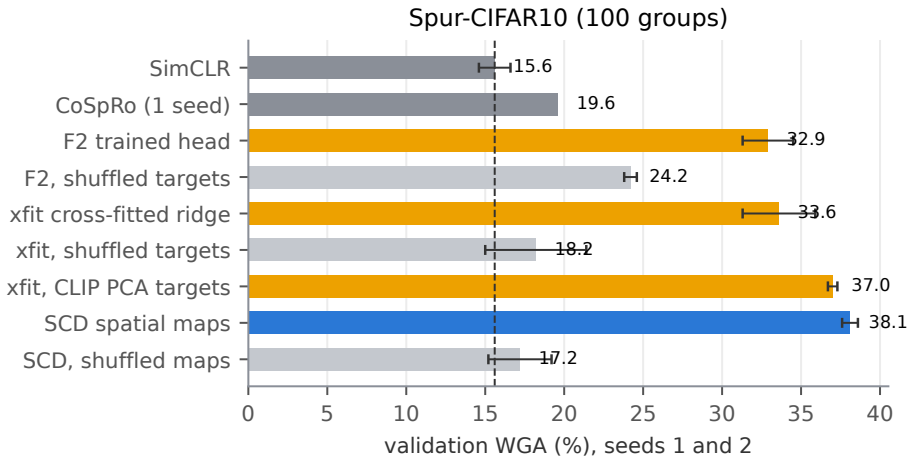


Figure 13: Spur-CIFAR10, where the spurious feature is a thin coloured line. Every real-target method roughly doubles WGA and every shuffled control stays near SimCLR. SCD gives the best result of the branch.

The advantage holds across training (Figure 14). On MetaShift the smoothed mean WGA of SCD leads all other methods at every probe from epoch 100 onward; from epoch 200 its margin over the closest method ranges from 3.4 to 8.0 points. On Waterbirds SCD leads SimCLR and the shuffled control from epoch 100 onward and the principal-component maps follow the same path. Average accuracy separates earlier: SCD leads all methods on both datasets from epoch 125 onward.

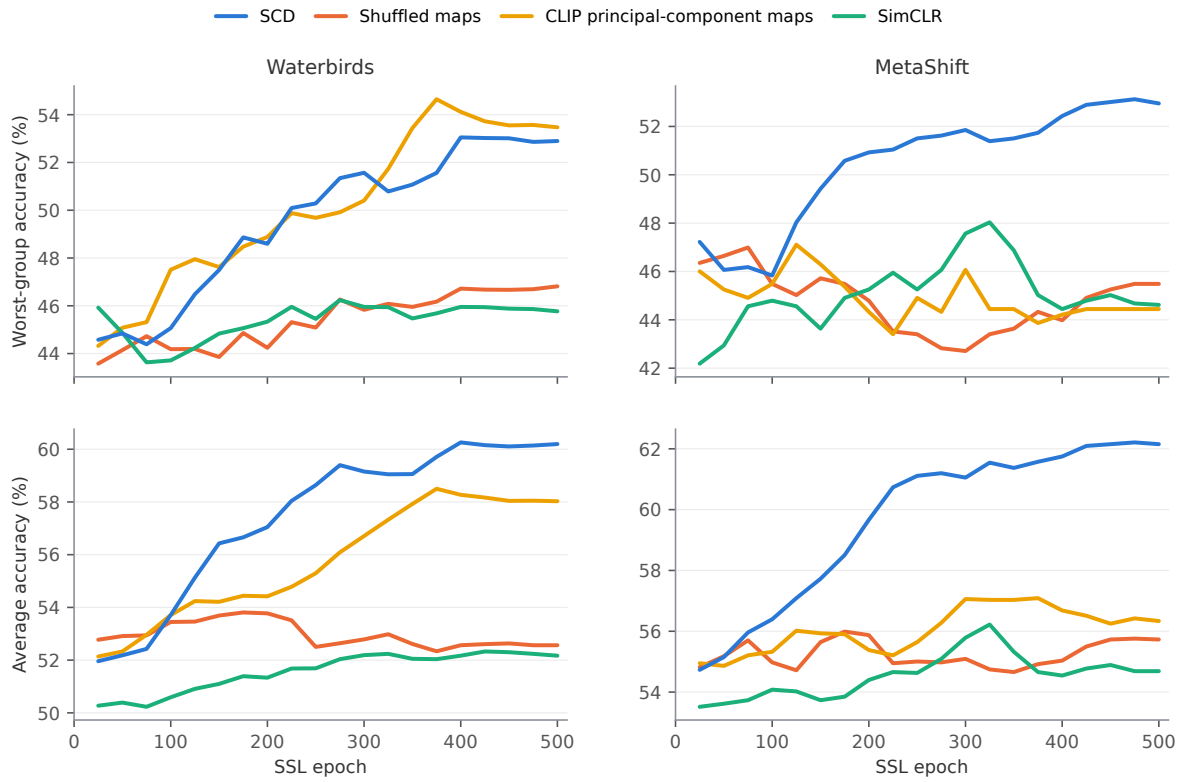


Figure 14: Validation accuracy during SSL training (seeds 1–4, 3-point smoothing). Top: WGA; bottom: average accuracy.

7.7 Controls: concepts, CLIP or location?

Three controls separate the ingredients of SCD. The principal-component control replaces the concept maps by as many principal components of the dense ClearCLIP patch embeddings as there are factors, computed over the same cells and trained with the same loss: it keeps the dense CLIP information of the teacher and removes the concept names. The image-level control regresses each image's SpLiCE factor activations from the pooled features with the same cross-fitted ridge and weight: it keeps the concepts and removes their locations. A concept ablation removes groups of factors.

Table 8: Controls of SCD, seeds 1–4 (image-level row on MetaShift: seeds 1–2).

	Waterbirds		MetaShift	
Method	Avg.	WGA	Avg.	WGA
SimCLR	52.3	45.8 ± 1.7	54.8	44.8 ± 9.4
Shuffled maps	52.6	46.7 ± 2.7	55.6	45.3 ± 3.4
CLIP principal-component maps	58.1	53.6 ± 1.9	56.3	44.4 ± 7.4
Image-level factor regression	58.0	50.4 ± 9.0	61.7	47.2 ± 3.9
SCD	60.2	53.0 ± 3.0	62.2	53.0 ± 5.8

- **Principal-component maps.** On MetaShift SCD wins on all four seeds (+16.0, +1.4, +4.2, +12.9); the PCA maps stay at the SimCLR level. On Waterbirds the two tie: birds against land and water are among the strongest directions of CLIP's patch space, so the leading components carry them; cats against dogs and couches against lawns are weaker directions that only the named concepts single out.
- **Image-level regression.** SCD is ahead by 2.7 WGA on Waterbirds, 8.3 on the two paired

MetaShift seeds and 8.7 on Spur-CIFAR10 (29.4 against 38.1).

- **Concept ablation** on MetaShift (seeds 1–4), an analysis with word lists we chose, outside the method: without the 21 animal factors 46.6 ± 5.8, without the 20 scene factors 47.9 ± 2.8, all factors 53.0 ± 5.8. Both ablations lose most of the gain: it needs the full concept set.

7.8 What the student learns

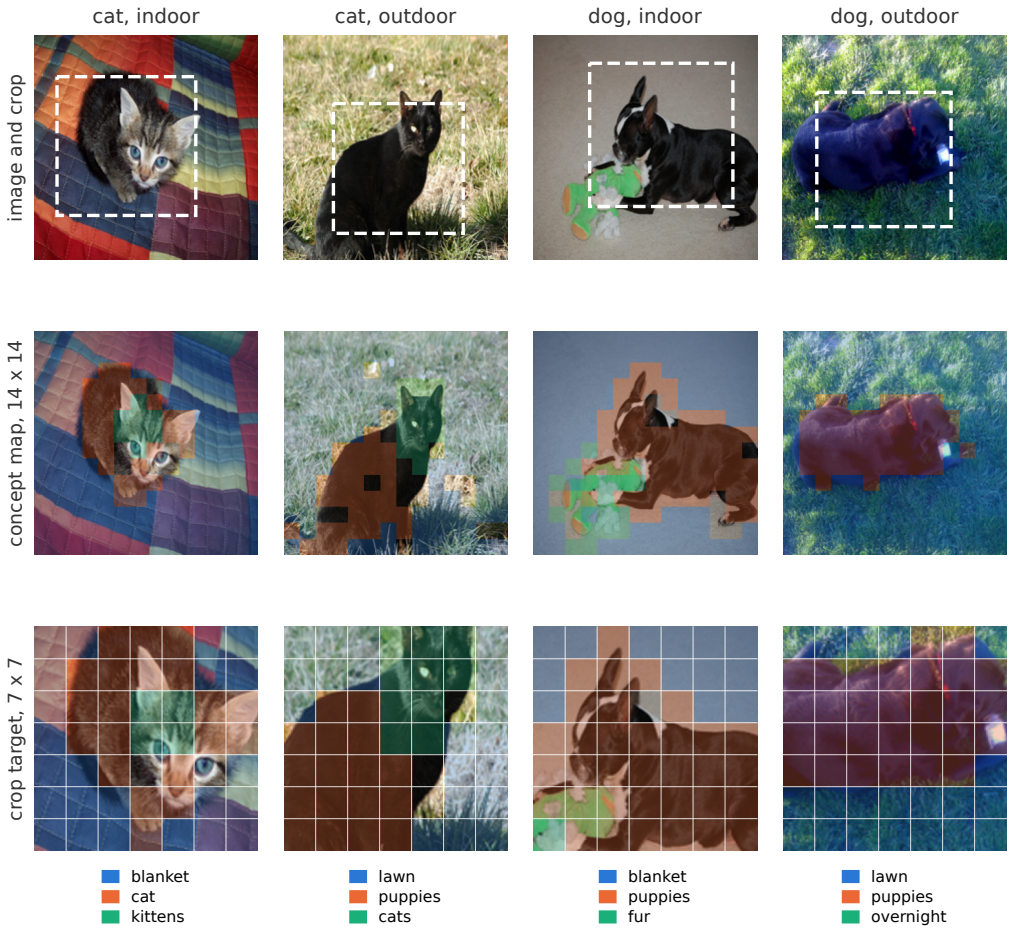


Figure 15: MetaShift concept maps. Top: image and one SimCLR crop (dashed). Middle: the winning factor of each of the 14×14 cells (the three most frequent factors coloured). Bottom: the crop's 7×7 targets after warping. The animal factors cover the animal, the scene factors its surroundings. The teacher is imperfect: in the cat-outdoor image part of the black cat is labelled "puppies".

What the maps contain. The MetaShift factors include scene concepts such as couch, lawn, beds, windows and bench next to the animal concepts; blankets, lawn, cats and fields win the most map cells (7.4%, 6.6%, 5.8% and 5.6%). On Waterbirds bamboo, forest and tree stand win 37% of the cells and the factor *water bird* 6.2%.

The student generalizes. On the held-out fifth the student's pooled features explain 0.22 (MetaShift) and 0.26 (Waterbirds) of the image-level factor responses, against 0.13 and 0.17 with shuffled maps; scores on images inside the loss agree (0.23 and 0.26). The memorization of the image-level line is gone (Figure 9).

The student keeps the co-occurrence. With labels used only for this audit, the student's *cat* factor scores 0.47 on unseen cats indoors and 0.07 on unseen cats outdoors; CLIP scores 0.92 and 0.88 (Figure 16). The factor residuals score 0.00 in the student against 0.15 to 0.17 for CLIP. A background probe on the frozen features reaches 90.5% WGA for SCD against 89.5% for SimCLR:

SCD keeps the background as readable as before. Correlated generative factors in the training data persist in learned representations (Träuble et al., 2021) and each 7×7 location sees most of the view (Section 4.2); the cheapest predictor of "cat here" therefore combines local evidence with the scene, which agrees with the cat on 88% of the training images.

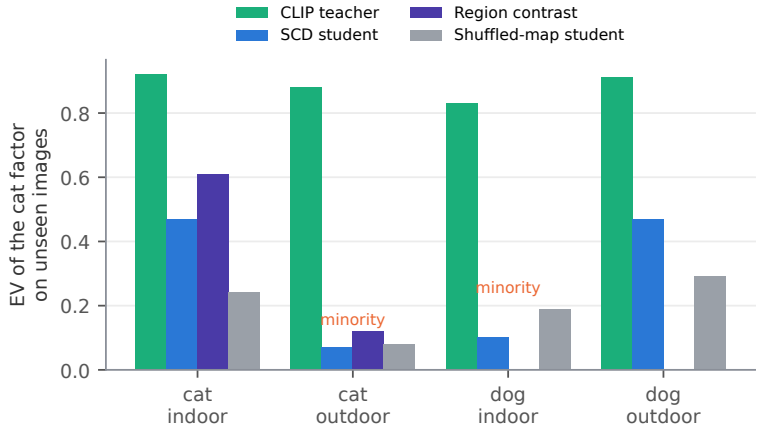


Figure 16: How well the *cat* factor is predicted on unseen MetaShift images, per group. CLIP sees cats everywhere; every student sees cats where cat and scene agree (cat indoors) and fails on the minority groups.

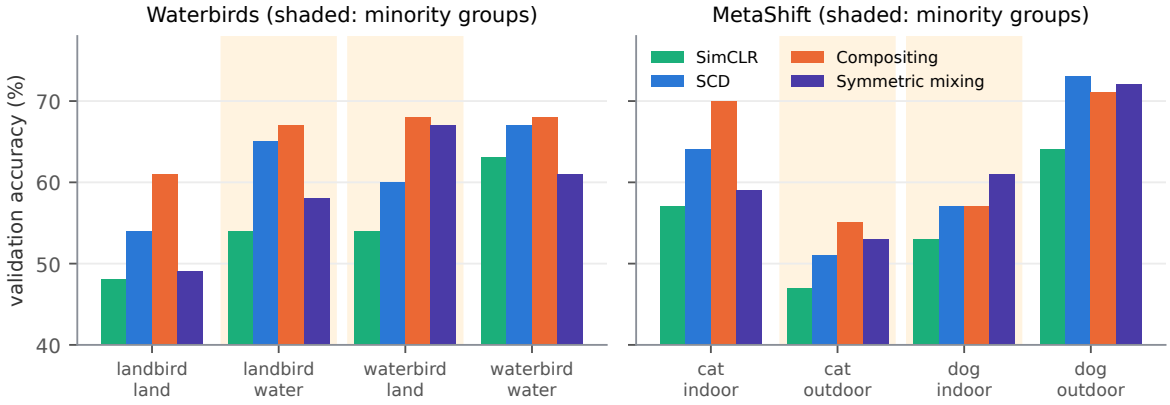


Figure 17: Accuracy per group (six seeds for SimCLR and SCD; compositing six seeds on Waterbirds, four on MetaShift; mixing four). SCD raises every group, the minority groups included; compositing flattens the Waterbirds groups at 61 to 68.

7.9 Limitations of SCD

All results are validation results with the probe evaluated on the same split; the locked test runs are prepared. The weight $\lambda_{\max} = 5$ and the hold-out were chosen on the validation splits of MetaShift and Waterbirds with seeds 1–4; seeds 5 and 6 ran with the settings fixed. MetaShift's seed spread (5.4 WGA points) is as large as the gain over its controls on single seeds. On Waterbirds dense CLIP components match the concept maps, so the concept names add value only where the relevant contrast is a weak direction of CLIP. The method depends on CLIP's external pretraining, its vocabulary and its 32-pixel patches: Spur-CIFAR10 images are upsampled from 32 to 448 pixels for the maps and CelebA has no concept maps yet.

7.10 Conclusion on SCD

SCD uses dense maps of SpLiCE concepts to supervise every location of a SimCLR student through a regression cross-fitted between halves of the batch, with no labels and no concept type during pretraining. On Waterbirds, MetaShift and Spur-CIFAR10 its mean validation worst-group and average accuracies exceed those of SimCLR and of the same loss on shuffled maps. On MetaShift the named concepts beat dense CLIP components of the same width; on Waterbirds the two tie. SCD raises WGA because the balanced probe gains a usable core feature that SimCLR from scratch lacks, while the student's concepts still follow the training co-occurrence. SCD is the prior-free method of the thesis.

8 Separating object and context

Every loss so far is an expectation over the natural training images, where cats sit indoors. The four extensions of this section either change that distribution or demand that a concept keeps its identity in other surroundings. All four run on top of SCD (weight 5, 20% hold-out). The theoretical motivation is the content–style result of von Kügelgen et al. (2021): contrastive learning isolates the content that its augmentations keep fixed, provided the augmentations change the style independently of it. Natural crops never move a cat outdoors, so the scene stays part of the "content".

8.1 Region contrast (DetCon-style)

Summary. The feature map is pooled inside each view's dominant concept regions and the pooled regions are contrasted across the batch, so that a cat region on a lawn must resemble a cat region on a couch. The concept scores on unseen images rise on every seed, yet WGA moves within noise (+2.3 on Waterbirds, −3.0 on MetaShift against SCD on the same seeds) and the gain goes to the majority groups. Region pooling inherits the receptive-field problem of Section 4.2: a region feature still sees the scene and on Waterbirds a shuffled region arm gains as much as the real one.

Method. Following DetCon (Hénaff et al., 2021), each view's feature map is pooled inside its two dominant concept regions (warped map probability as weight, mean probability at least 0.1), projected with a small head and contrasted across the batch: regions of the same concept in other images are positives, weighted towards pairs whose images' concept mixes differ. About 335 regions form per MetaShift batch, 94% of them with positives.

Result. Waterbirds 55.3 ± 3.1 (SCD 53.0 on the same seeds), MetaShift 50.0 ± 6.5 (below SCD on three of four seeds). The concept scores on unseen images rise on every seed (MetaShift 0.22 → 0.29) and the cat factor on cats outdoors from 0.05 to 0.12. On Waterbirds the shuffled region arm gains as much over shuffled SCD as the real one over SCD: part of the gain is region pooling itself.

8.2 Concept compositing (with the object prior)

Summary. The object region of one image is cut out and pasted onto the context of another image, whose own objects are painted over; the composite replaces the second SimCLR view. With a random donor, object and context become independent in the composites. It works best of all variants: Waterbirds 59.1 WGA, ahead of SCD on all six seeds, and the steadiest MetaShift result (52.3, sd 1.9). It needs a text prior that types each concept as an object or a scene, which the restriction excludes; compositing therefore measures what a correct prior adds.

Method. For 30% of the qualifying images, the *object* region of the second view is cut out and pasted onto another image's view whose own objects are painted over by normalized convolution from their surroundings (Knutsson and Westin, 1993). Cut-and-paste synthesis is an established augmentation for detection and segmentation (Dwibedi et al., 2017; Ghiasi et al., 2021); DiLo and object-aware contrastive learning paste salient objects onto other backgrounds during SSL (Mo

et al., 2021; Zhao et al., 2021). The concept targets are composited with the same mask. With a random donor, the composites sample the interventional distribution in which the context carries no information about the object (Pearl, 2009); augmentations that simulate such interventions yield invariance to the intervened factor (Ilse et al., 2021; Mitrovic et al., 2021), which is the content–style condition of von Kügelgen et al. (2021).

Which concepts are "objects" is decided by text alone (`cospro/pipeline/concept_type.py`): CLIP text similarity to scene prompts minus similarity to object prompts; factors below the median count as objects. No label enters, but the step *types* concepts. On MetaShift the 65 object factors hold every animal and breed (and also verbs and people); lawn, couch, bed, desk, window, beach and snow are context.

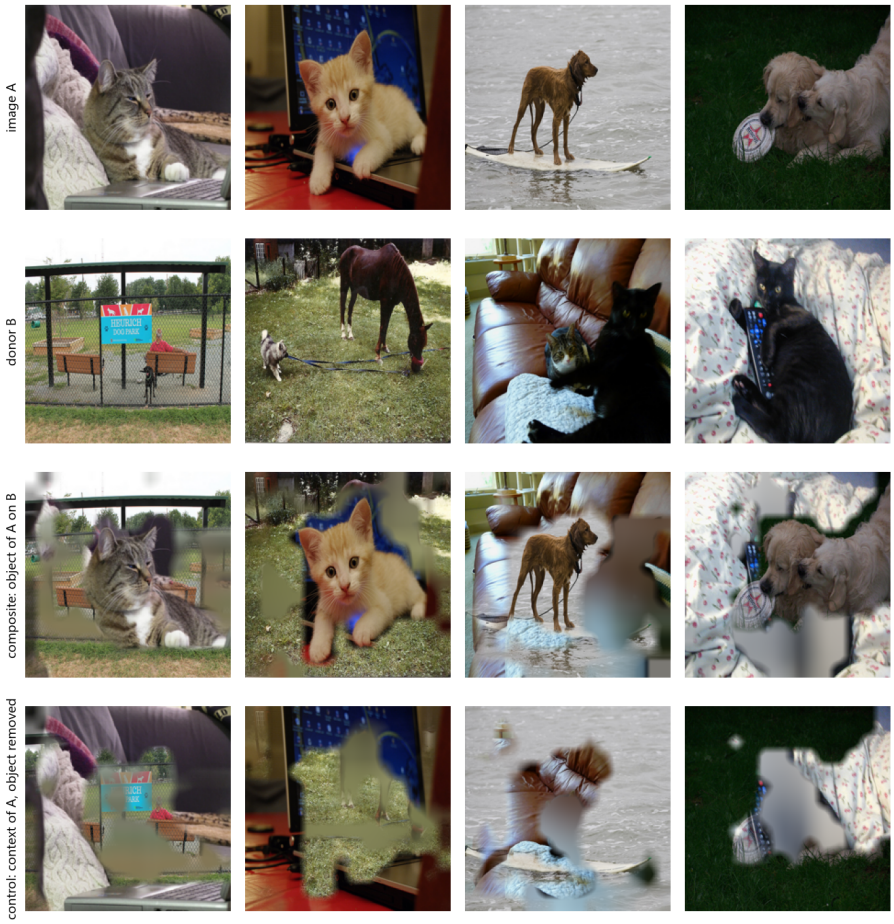


Figure 18: Real MetaShift composites. Row 3: the object of image A on donor B (the training input). Row 4: the "context kept" control, which pastes A's surroundings instead. Seams are half a map cell wide.



Figure 19: The object mass that compositing cuts (row 2) and the three leading concepts per cell (row 3) on one MetaShift image per group. The teacher often names the context well (windshield, driving, lawn) and the animal in broad strokes.

Table 9: Compositing and its controls (validation WGA / Avg.). Waterbirds seeds 1–6 for compositing and the silhouette control, 1–4 otherwise; MetaShift seeds 1–4.

Arm	Waterbirds	MetaShift
SCD	52.8 / 60.3	51.1 / 61.2
Compositing, object regions	59.1 ± 7.1 / 64.7	52.3 ± 1.9 / 63.3
control: another image's object silhouette	54.2 ± 6.1 / 62.5	50.6 ± 3.7 / 60.0
control: context region pasted instead	52.7 ± 2.1 / 60.0	49.3 ± 1.9 / 60.1
control: shuffled maps	52.9 ± 1.0 / 57.2	48.4 ± 3.1 / 56.6

Result. On Waterbirds compositing beats SCD on all six seeds (6.3 WGA on average), the context and shuffled controls on all four paired seeds and the silhouette control on four of six; its groups flatten to 61–68. On MetaShift it gives the steadiest WGA of the branch (sd 1.9). The arms order as the concept hypothesis predicts: object regions > random silhouettes > context regions. The gain rests on the object prior (target = object, spurious attribute = surroundings), which the restriction excludes and which CelebA breaks. Background augmentations report the same kind of gain under the same prior (Ryali et al., 2021).

8.3 Symmetric concept mixing (no prior)

Summary. The prior-free version of compositing: every concept region of an image is kept or replaced at random, so no concept is typed. It falls below SCD on Waterbirds (49.2), ties it on MetaShift (51.0) and ties its own control with random rectangles on both datasets. Without the object type the method reduces to CutMix-like mixing: half of the mixes keep the scene and replace the cat, which teaches the scene as much as the cat.

Method. Each concept region of view A is kept with probability one half (kept area 25–75%) and

the rest filled from a random donor B: $x = M \odot A + (1 - M) \odot B$, as in CutMix (Yun et al., 2019) with concept-shaped masks. The mixed view is a third encoder pass. Its feature map pooled inside M is a positive of A and pooled outside a positive of B, as in RegionCL (Xu et al., 2022); its global embedding is a soft positive of A and B in proportion to their areas, as in Un-Mix (Shen et al., 2022). Under mixing the twelve strongest animal-by-scene correlations on MetaShift fall from mean $|\phi| = 0.33$ to 0.19 (object pasting: 0.17).

Result. Waterbirds 49.2 ± 2.5 (below SCD on all four seeds), MetaShift 51.0 ± 8.5; it ties its rectangle control on both datasets (49.0 and 51.8), so concept boundaries make no difference. The mixing weakens the majority group landbird-on-land (49 vs. 55).

8.4 Per-group projection heads (LooC-style, no prior)

Summary. Symmetric mixing asks one embedding to stay the same whatever part was replaced, sometimes the scene and sometimes the cat and these demands may cancel. Separate projection heads per concept group remove the conflict. The heads train well (the highest unseen concept scores of all arms), but the effect changes sign between datasets: Waterbirds 52.4 (ties SCD), MetaShift 46.9 (below SCD by 6.2). The class groups are kept in only 21% of the mixes, and drawing them more often would require knowing the class, which is the prior again. This ended the prior-free mixing line.

Method. LooC gives each invariance its own embedding space (Xiao et al., 2021b). Factors are clustered into 12 groups by the pixels they share in the maps (spectral clustering of map co-activation, Ng et al., 2001; no label or prompt). For half of the images one group is kept on a donor; the mix is a positive of its image only in that group's own projection head.

Result. Waterbirds 52.4 ± 2.9 (ties SCD, beats the single-head control by 9.5); MetaShift 46.9 ± 3.3 (below the single head by 5.2, below SCD by 6.2 and below its shuffled control). Concept scores on unseen images (0.28 and 0.32) do not predict WGA.

9 Comparison of all architectures

Table 10: All architectures with their idea, outcome and the lesson they taught. WGA on Waterbirds / MetaShift (validation).

Architecture	Idea	WGA (WB / MS)	Lesson
F2 trained head	regress factor values from the pooled feature	41.6 / 46.6	per-image targets are memorized on small datasets
F3 concept blocks	SupCon per factor, cross-context positives	– / 43.8	shuffled concept sets fit as well
xfit	ridge fitted on one half, tested on the other	51.5 / 47.6	removes head memorization; backbone still memorizes; CLIP PCA as good
SCD	per-location cross-fit on dense concept maps	**52.8 / 51.1**	location helps; generalizes; keeps the co-occurrence
Region contrast	DetCon over concept regions	55.3 / 50.0	raises concept scores, minority groups stay
Compositing	paste object on another scene	**59.1 / 52.3**	works, needs object-vs-scene prior
Symmetric mixing	keep random concept regions, soft positives	49.2 / 51.0	equals CutMix with rectangles
Per-group heads	one embedding space per concept group	52.4 / 46.9	sign flips; drawing the class needs the prior

10 Interpretation

1. **Concepts help a from-scratch SSL student when they are delivered where they are.** Image-level distillation is limited by memorization and by co-occurrence; spatial distillation generalizes and lifts every group.
2. **Every claim stands on a matched control.** Shuffled targets test content, PCA maps test named concepts against generic CLIP information, image-level regression tests location and ablations test which concepts matter.
3. **The core feature becomes stronger and stays fused with its context.** DFR needs separate directions; SCD supplies a stronger core direction that still carries the scene. Loosening the fusion worked only with an object prior.
4. **The decision for the thesis.** The prior-free method is SCD. Compositing reports what a correct assumption about the spurious attribute adds, stated openly, as the background-augmentation literature does (Ryali et al., 2021).

11 Limitations and next steps

- **Validation only.** Model selection on the evaluation split inflates results (Gulrajani and Lopez-Paz, 2021). `FINAL_TEST=1 bash scripts/submit_factor_study.sh <dataset>` runs the same arms once on the locked test split (Waterbirds test: 642 images per minority group against 133 in validation).
- **Noise.** MetaShift's seed spread (5.4 WGA) is as large as the gain on single seeds.
- **The hold-out effect** (higher WGA with 20% of images out of the concept loss) has no explanation.
- **LateTVG matched comparison**: one seed under our protocol (47.9); the six-seed test run is predeclared.
- **CelebA**: concept maps for 162,770 images exceed the loader's memory as float32 copies; image-level xfit gives 85.9 against 84.9 for SimCLR.
- **Teacher limits**: CLIP's vocabulary and 32-pixel patches; Spur-CIFAR10 images are upsampled from 32 to 448 pixels for the maps.

Proposed next steps (to agree on in the meeting):

1. Locked test for SimCLR, shuffled maps, SCD and compositing on Waterbirds and MetaShift, six seeds.
2. Matched LateTVG on Waterbirds (predeclared) and MetaShift (prune rate chosen on validation first).
3. CelebA concept maps with a memory-lean loader, then SCD on CelebA as the held-out dataset whose spurious attribute lies in no background.
4. Decide how the thesis frames compositing: as an ablation that quantifies the object prior.

12 Questions the supervisor may ask

Q: Why train from scratch and use ResNet-18 instead of a pretrained model?

A: The protocol is inherited from CoSpRo and follows LateTVG and LA-SSL, which pretrain on the dataset's own images, so that every method sees the same images and budget. A pretrained

backbone already encodes birds and cats from ImageNet and would mask what the SSL objective learns from the dataset (Izmailov et al., 2022). The cost is low absolute accuracy (52–61%).

Q: Is this just CLIP distillation?

A: At the image level it is: CLIP principal components match the concepts. At the spatial level the concept maps beat PCA maps of the same width on MetaShift on all four seeds and tie on Waterbirds, where bird-versus-background is already a leading CLIP direction.

Q: Where do labels enter?

A: Only in the group-balanced probe (224 / 400 / 1,400 images) and the post-hoc audit, identically for every method. Concept discovery, grouping, maps and SSL use none. Hyperparameters were chosen on validation of MetaShift and Waterbirds, which is the main caveat.

Q: Why not use the pooled feature for the concept loss?

A: Pooling fuses co-occurring concepts into one vector; at 7×7 they occupy different locations (Section 4.2).

Q: Why does SCD leave cat and indoor fused?

A: Each 7×7 location's receptive field covers most of the view and the scene predicts the cat on 88% of the training images, so the cheapest predictor of "cat here" uses the scene. The loss localizes concepts and keeps the training joint distribution.

Q: Why is compositing not the main method?

A: It needs a text prior that types concepts as objects or scenes. That prior holds for Waterbirds and MetaShift and fails for CelebA; the thesis restriction excludes it.

Q: How do you know the gain is real?

A: Six seeds, paired differences (SCD ahead on five of six on both datasets) and controls that share every statistic but the content, which stay at the SimCLR level.

Q: Why the last four probes?

A: One MetaShift validation image is 1.4 points of a group's accuracy; averaging the probes at epochs 425–500 damps that noise. For SCD, SimCLR and compositing the final-probe numbers differ from these means by less than one point.

Q: Why did F2 work on Spur-CIFAR10 and fail on MetaShift?

A: 45,000 images resist memorization (shuffled targets reach 0.02 explained variance), 1,700 images do not (0.33).

A Arm name decoder

Arm	Meaning
`simclr`	plain SimCLR; `simclr_lt`: LateTVG's SSL hyperparameters
`f2_std`, `f2_w3`, `f2_w10`	F2 trained head, standardized targets, weight 1 / 3 / 10
`f2_white*`, `*_atyp`, `*_balanced`	whitened targets; atypicality weights; balancing weights
`cbc`, `cbc_plain`	F3 concept blocks; with uniform positive weights
`xfit`, `xfit_norm`	cross-fit; `_norm` = unit rows, absolute ridge
`xfit_meaning_w10_holdout`	meaning factors from 5%, weight 10, 20% hold-out
`*_clippca`, `*_response`, `*_residual`	targets: CLIP principal components; dense image responses; residuals
`spatial_w5_holdout`	**SCD** (spatial weight 5, 20% hold-out); `spatial_w5`: without hold-out
`spatial_pca_w5_holdout`	PCA-map control; `_noanimals`, `_noscenes`: concept ablations
`region_w1_holdout`	SCD + region contrast weight 1
`comp_w5_holdout`	SCD + compositing; `comp_random`, `comp_context`: silhouette and context controls
`mix_w5_holdout`	symmetric mixing; `mix_box`: rectangles; `mix_hard`: hard positive; `mix_noregion`: no region term
`heads_w5_holdout`	per-group heads; `heads_single`: one head; `heads_random`: random head
`*_shuffled*`	same loss on another image's targets or maps

B Per-seed and per-group results of SCD

Table 11: Waterbirds validation, mean of the last four probes. Groups (y, a): landbird on land, landbird on water, waterbird on land, waterbird on water.

Method	Seed	Avg.	WGA	(0,0)	(0,1)	(1,0)	(1,1)
SimCLR	1	50.67	44.97	44.54	50.00	54.89	66.92
SimCLR	2	52.00	43.79	43.68	58.37	52.63	57.89
SimCLR	3	52.90	47.43	47.32	56.22	51.88	59.40
SimCLR	4	53.44	46.99	53.75	50.43	46.62	70.68
SimCLR	5	52.15	48.18	47.54	50.86	58.65	63.16
SimCLR	6	53.48	48.61	48.61	55.58	57.89	60.15
Shuffled maps	1	56.46	50.19	53.32	60.52	50.38	57.89
Shuffled maps	2	53.29	47.54	47.75	56.87	59.40	54.14
Shuffled maps	3	52.61	44.91	45.61	59.44	51.88	55.64
Shuffled maps	4	48.02	44.27	43.90	45.28	55.64	60.15
Shuffled maps	5	53.19	48.07	48.18	56.22	54.14	58.65
Shuffled maps	6	53.17	50.00	50.75	56.22	51.13	54.14
SCD	1	59.97	57.26	58.03	58.80	57.89	74.44
SCD	2	63.45	51.13	58.24	72.96	51.13	60.90
SCD	3	59.70	50.75	50.54	63.73	71.43	68.42
SCD	4	57.49	52.96	52.68	60.09	54.89	68.42
SCD	5	63.55	58.65	58.67	70.60	59.40	58.65
SCD	6	57.78	45.82	45.61	64.38	64.66	73.68

Table 12: MetaShift validation, mean of the last four probes. Groups (y, a): cat indoors, cat outdoors, dog indoors, dog outdoors.

Method	Seed	Avg.	WGA	(0,0)	(0,1)	(1,0)	(1,1)
SimCLR	1	56.34	50.69	59.72	50.00	58.33	55.56
SimCLR	2	55.30	44.79	50.00	62.50	44.44	63.89
SimCLR	3	48.09	31.60	48.61	30.56	50.00	61.11
SimCLR	4	59.46	52.08	61.11	51.39	58.33	66.67
SimCLR	5	53.82	44.44	62.50	45.83	44.44	63.89
SimCLR	6	57.99	42.36	58.33	41.67	61.11	70.83
Shuffled maps	1	58.07	43.06	69.44	45.83	43.06	72.22
Shuffled maps	2	54.25	42.01	48.61	43.06	52.78	73.61
Shuffled maps	3	56.94	49.31	62.50	48.61	50.00	65.28
Shuffled maps	4	53.21	46.87	62.50	47.22	47.22	52.78
Shuffled maps	5	58.85	46.53	62.50	45.83	48.61	77.78
Shuffled maps	6	55.12	42.71	43.06	43.06	55.56	77.78
SCD	1	66.06	60.42	69.44	61.11	68.06	66.67
SCD	2	58.85	50.69	59.72	51.39	51.39	75.00
SCD	3	61.63	54.17	62.50	52.78	58.33	70.83
SCD	4	62.33	46.87	69.44	47.22	55.56	79.17
SCD	5	60.33	48.61	63.89	48.61	56.94	73.61
SCD	6	57.73	45.83	58.33	45.83	52.78	72.22

C Where things live

What	Path
Image-level history (LaTeX)	`docs/xfit/xfit_history.tex`
SCD paper draft	`paper/SpatialConcepts.tex`
Method code (F2, xfit, SCD, region, compositing, mixing, heads)	`cospro/methods/concept_factors.py`
Concept maps (ClearCLIP)	`cospro/pipeline/concept_maps.py`, `scripts/build_concept_maps.sbatch`
Meaning grouping	`cospro/cli/build_meaning_groups.py`
Object/scene text scores	`cospro/pipeline/concept_type.py`
Study launcher (arms, at most 8 jobs)	`scripts/submit_factor_study.sh`
Result tables per dataset	`outputs/reports/factors_<dataset>/summary.md`

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